Botao Zhang

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Education

Fudan University *Shanghai, China*
B.S. in Mathematics, School of Mathematical Sciences Sept 2023 – Jul 2027 (expected)

Shanghai High School *Shanghai, China*
Sept 2020 – Jun 2023

Research Experience

Benna Lab, University of California, San Diego *San Diego, CA, USA*
Visiting Student · Advisor: Prof. Marcus Benna Jan 2026 – Present

- Developing a biologically plausible neural network that decomposes patterns into features, inspired by the similarity-matching approach in the SNMF algorithm from Cengiz Pehlevan's lab. This decomposition allows patterns to be represented sparsely as combinations of features, enabling efficient storage in a classical Hopfield network.

Laboratory of Cognitive Computational Neuroscience and Neuroimaging
Shanghai Jiao Tong University *Shanghai, China*
Visiting Student · Advisor: Prof. Ruyuan Zhang Oct 2023 – Dec 2025

- Investigated arbitration between model-free and model-based control in decision-making. Building on the reliability-based arbitration framework of Lee et al. (2014), developed and tested a simpler, equally intuitive arbitration model to explain behavioral data.

Technical Skills

Programming: Python, R, LaTeX
Languages: Mandarin Chinese (native), English (fluent)

Awards

First Prize, Chinese National High School Mathematics Olympiad, Shanghai 2021, 2022